# Neurobiological Mechanisms of Streptozotocin (STZ)-Induced Memory Impairment as a Reliable Animal Model of Alzheimer's Disease: A Systematic Review and Meta-Analysis Study

*Ameneh Rezayof, PhD,*

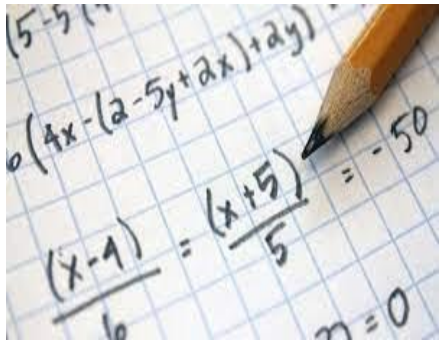
*Professor at the School of Biology, College of Science,*
*University of Tehran,*

*Tehran, Iran*

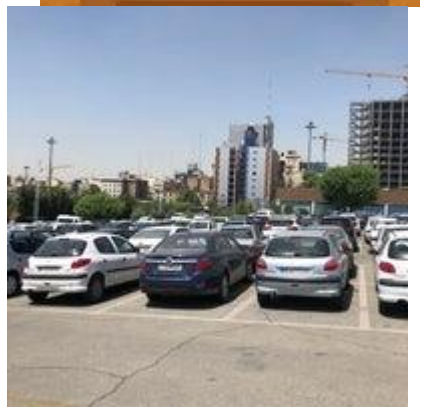

I wake up
I get up
I take a shower
I get dressed
I comb my hair
I have breakfast
I go to work
I start work
I answer emails
I have lunch
I finish work
I arrive home
I feed the dog
I cook dinner
I have dinner
I watch TV
I go to bed
I fall asleep








Good Memory
Is Important











# Memory formation is a complicated story

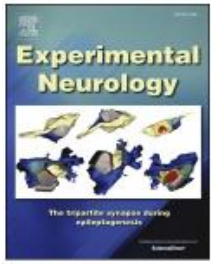


Experimental Neurology 370 (2023) 114560

Contents lists available at ScienceDirect

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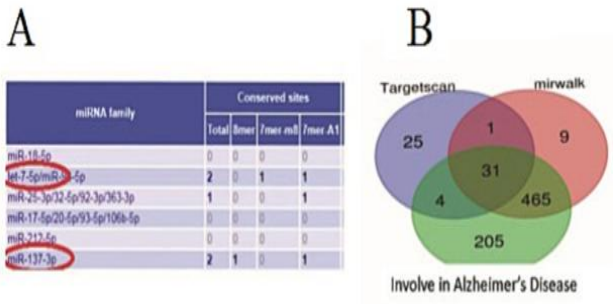
journal homepage: www.elsevier.com/locate/yexnr

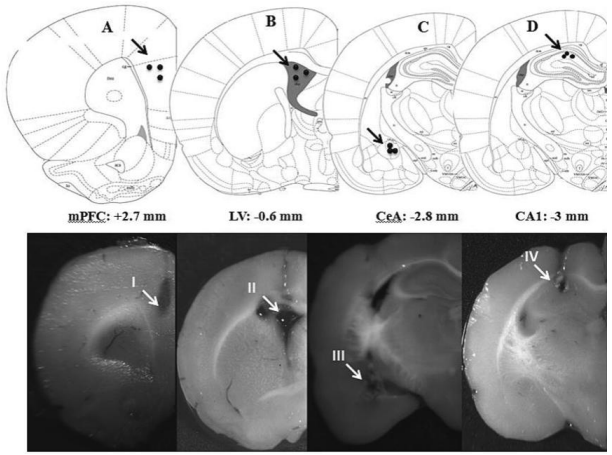


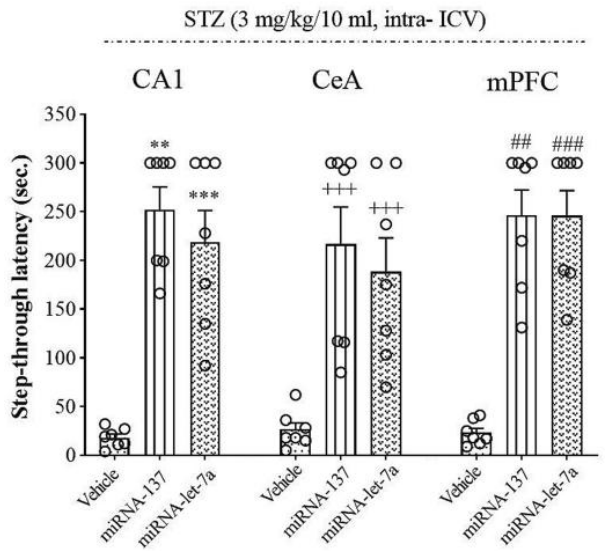

Research paper

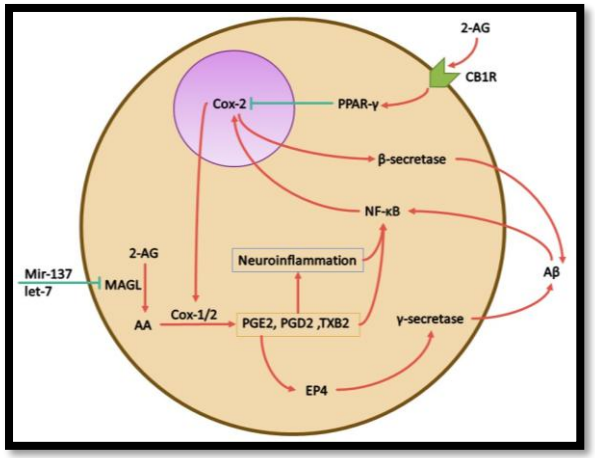
# Memory impairment was ameliorated by corticolimbic microinjections of arachidonylcyclopropylamide (ACPA) and miRNA-regulated lentiviral particles in a streptozotocin-induced Alzheimer's rat model

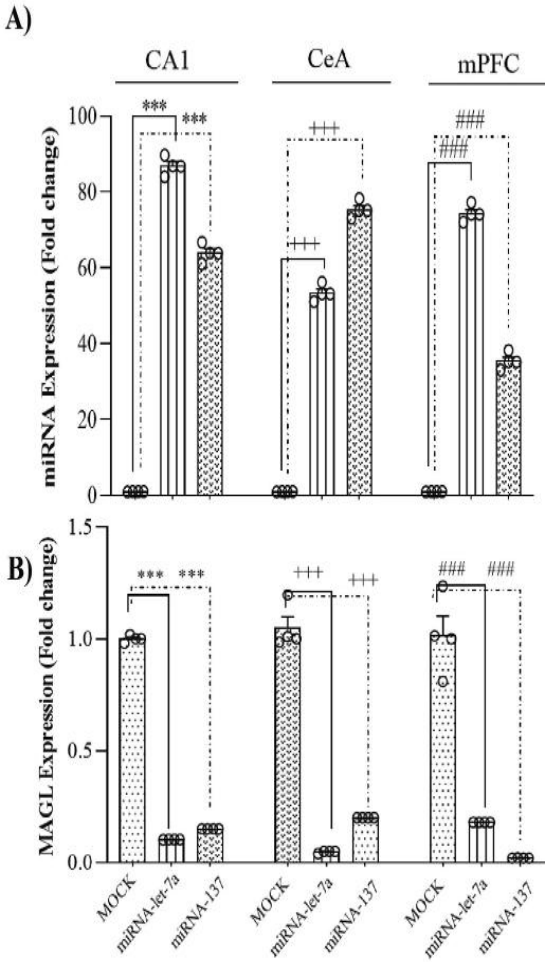
Mohammad Hosseininia [a], Fatemeh Rostami [b], Ladan Delphi [a], Zahra Ghasemzadeh [a], Fatemeh Kouhkan [b,*], Ameneh Rezayof [a,**]



# Animals

Wistar Male Rats

12hrs of light and darkeness

Temperature 22 ±2°C



200-220 gr

Free access To food and water

4 Rats per cage

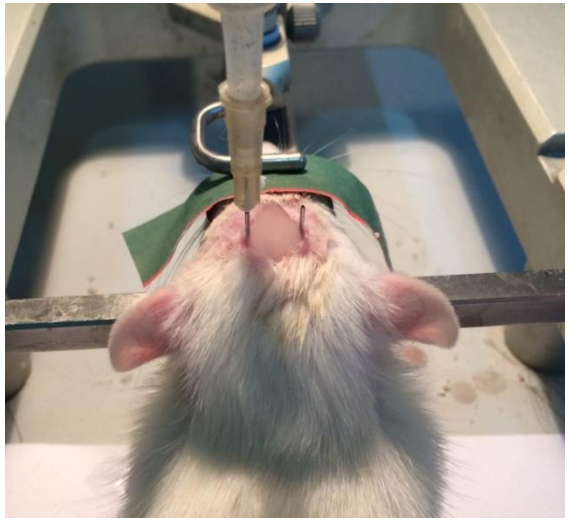
# Surgical Procedure and Drug injections



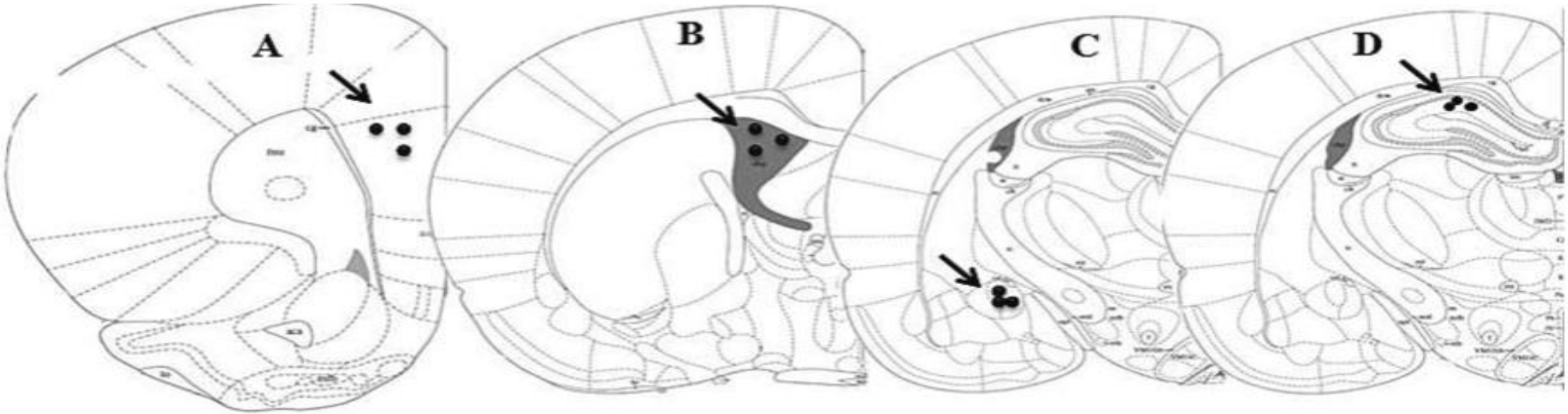




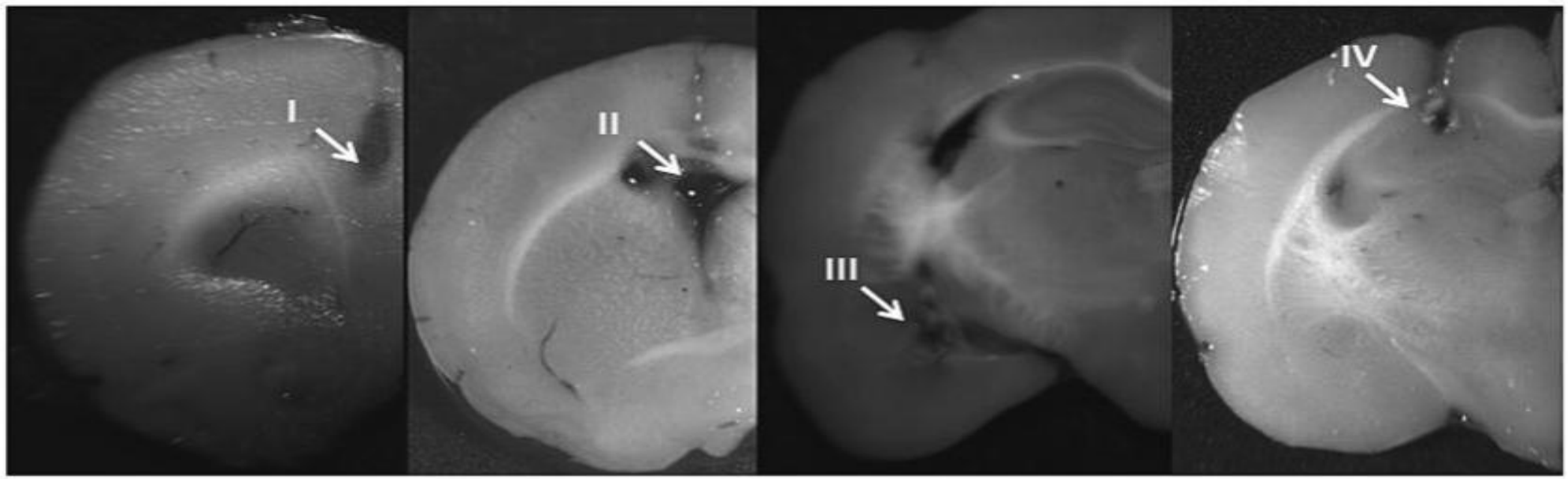






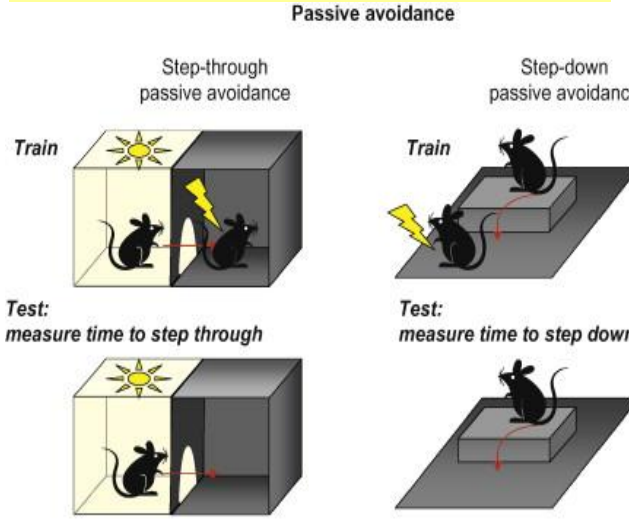



mPFC: +2.7 mm LV: -0.6 mm CeA: -2.8 mm CA1: -3 mm

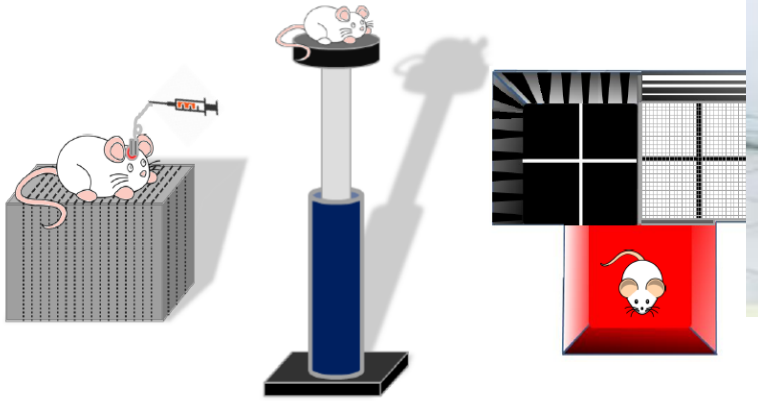


# Our Animal models

## Passive avoidance




## Conditioned place preference

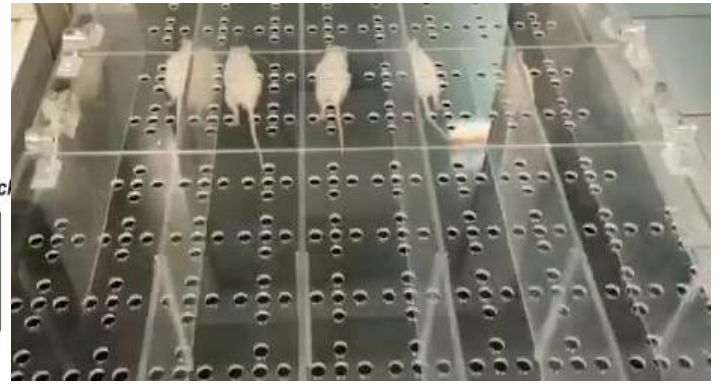




## Elevated plus maze

## Cognitive enhancers

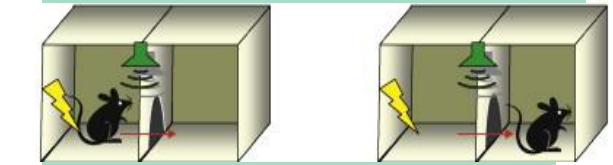
## Passive avoidance task

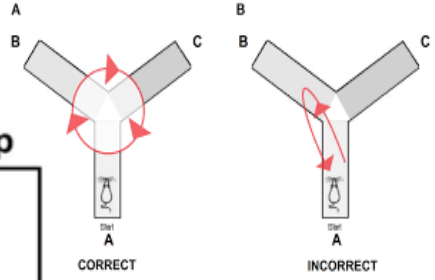




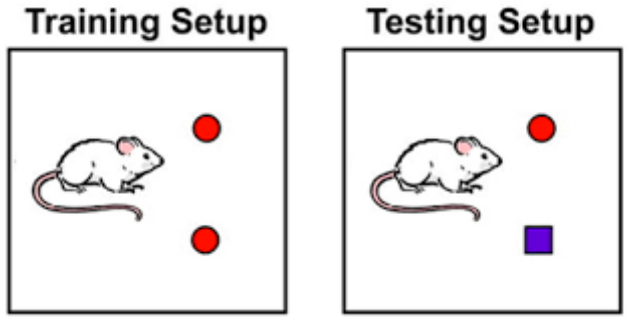
## Seeking Behaviors

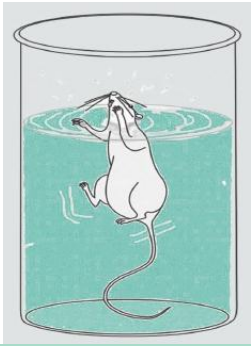
## Novel object recognition test






## Y maze



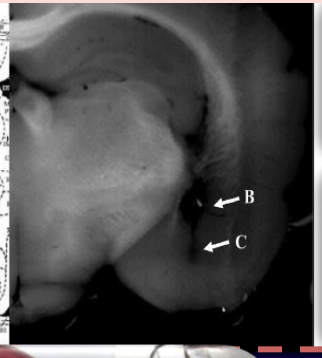
## Swimming test

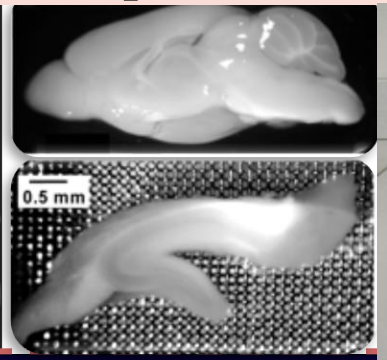


# Expression changes of proteins and microRNAs in the hippocampus, the amygdala and the prefrontal cortex

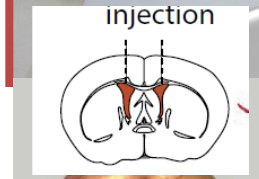


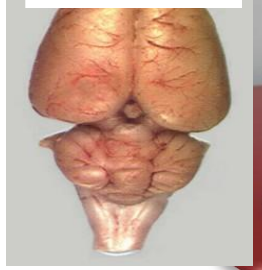




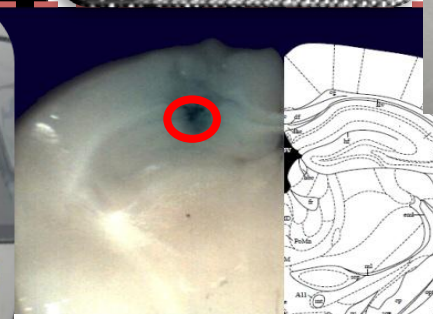


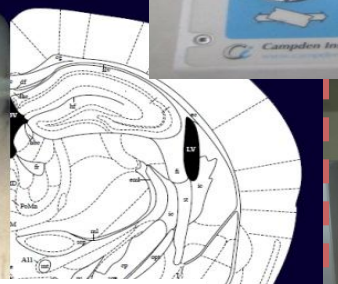





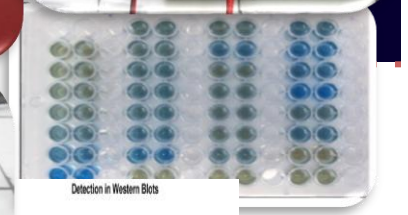


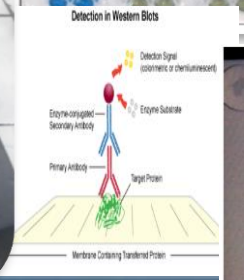


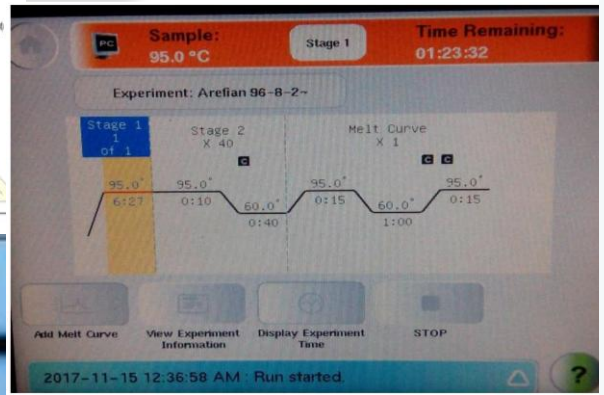











sample collection

Hip








*Bioinformatics* in *MicroRNA* Research

Tawfik *et al. Tropical Medicine and Health* (2019) 47:46
https://doi.org/10.1186/s41182-019-0165-6

Tropical Medicine
and Health

REVIEW Open Access

# A step by step guide for conducting a systematic review and meta-analysis with simulation data

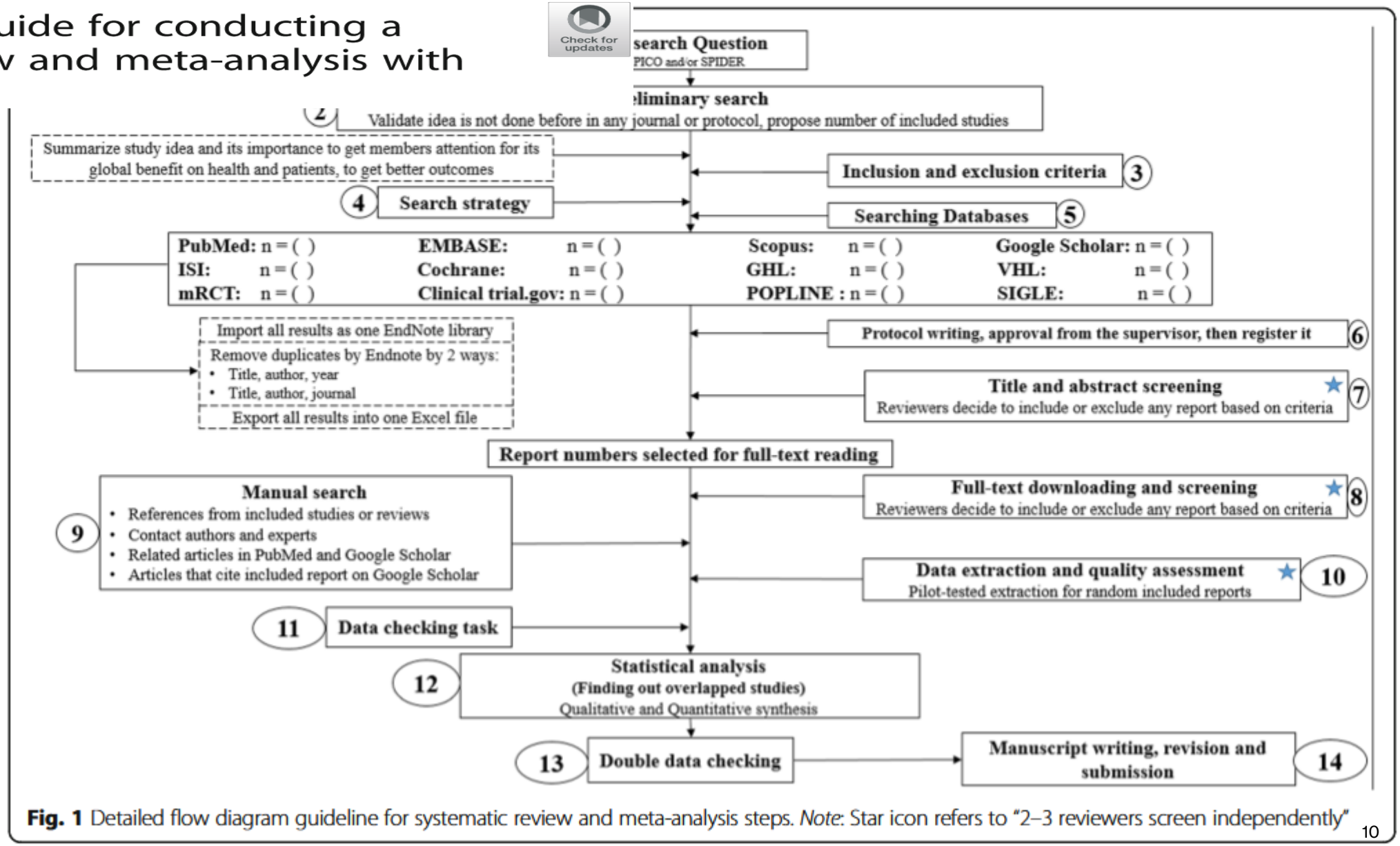
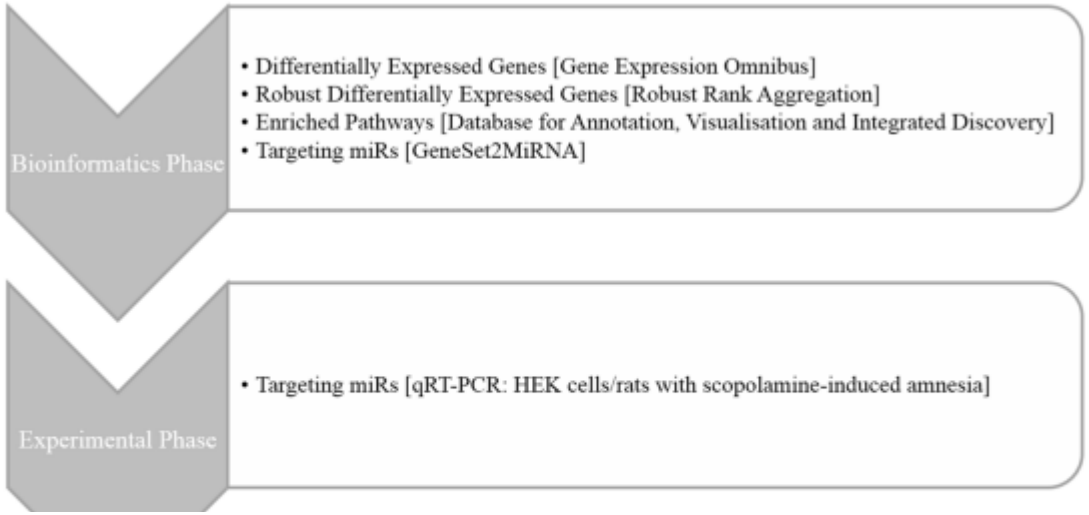


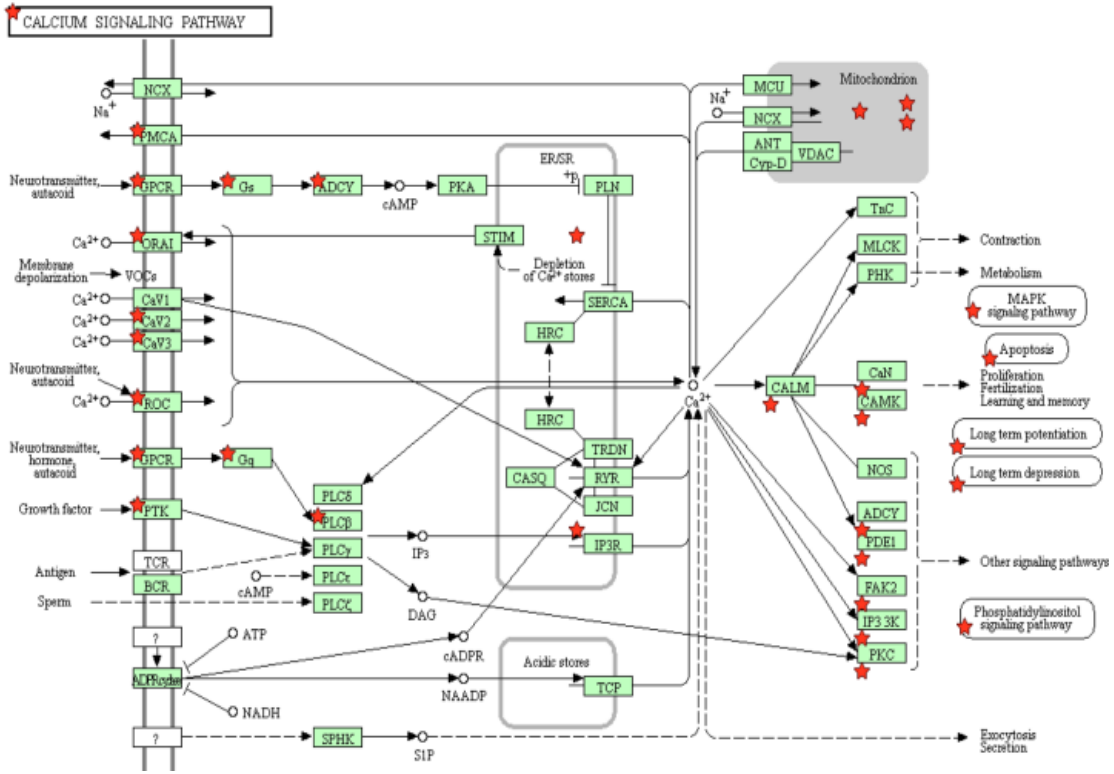

**Fig. 1** Detailed flow diagram guideline for systematic review and meta-analysis steps. *Note*: Star icon refers to "2–3 reviewers screen independently"

SCIENTIFIC REPORTS
nature research
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# A meta-analysis of gene expression data highlights synaptic dysfunction in the hippocampus of brains with Alzheimer's disease

Saeedeh Hosseinian[1], Ehsan Arefian[2,3]✉, Hassan Rakhsh-Khorshid[4], Mehdi Eivani[5], Ameneh Rezayof[6], Hamid Pezeshk[7,8] & Sayed-Amir Marashi[1]




| miR | Adj. *P*-value | Target Genes |
|---|---|---|
| hsa-miR-330-3p | 0.01 | *ATP2B1, ATP2B2, CHP, GRM5, PLCB1, PRKCB* |
| hsa-miR-129-5p | 0.01 | *ADCY2, ATP2B1, ATP2B3, CALM1, CAMK2D, CAMK4, PDGFRA, PPP3CA, PRKCB* |
| hsa-miR-614 | 0.01 | *ADCY1, CALM1, CAMK2A, GRM5, PLCB1* |
| hsa-miR-181b | 0.01 | *ADCY1, ATP2A2, ATP2B1, ATP2B2, CALM1, GRM5, PDGFRA* |
| hsa-miR-30c | 0.01 | *ADRB1, ATP2A2, ATP2B1, ATP2B2, CAMK2D, GRM5, PPID, PPP3CA* |
| hsa-miR-181d | 0.02 | *ADRB1, ATP2A2, ATP2B1, ATP2B2, CAMK2D, GRM5, PPID, PPP3CA* |
| hsa-miR-30a | 0.03 | *ATP2A2, ATP2B1, ATP2B2, CAMK2D, GRM5, PPID, PPP3CA* |

European Journal of Epidemiology (2020) 35:49–60
https://doi.org/10.1007/s10654-019-00576-5

GUIDELINES

# A 24-step guide on how to design, conduct, and successfully publish a systematic review and meta-analysis in medical research

Taulant Muka[1] · Marija Glisic[1,2] · Jelena Milic[3,4] · Sanne Verhoog[1] · Julia Bohlius[1] · Wichor Bramer[5] · Rajiv Chowdhury[6] · Oscar H. Franco[1]

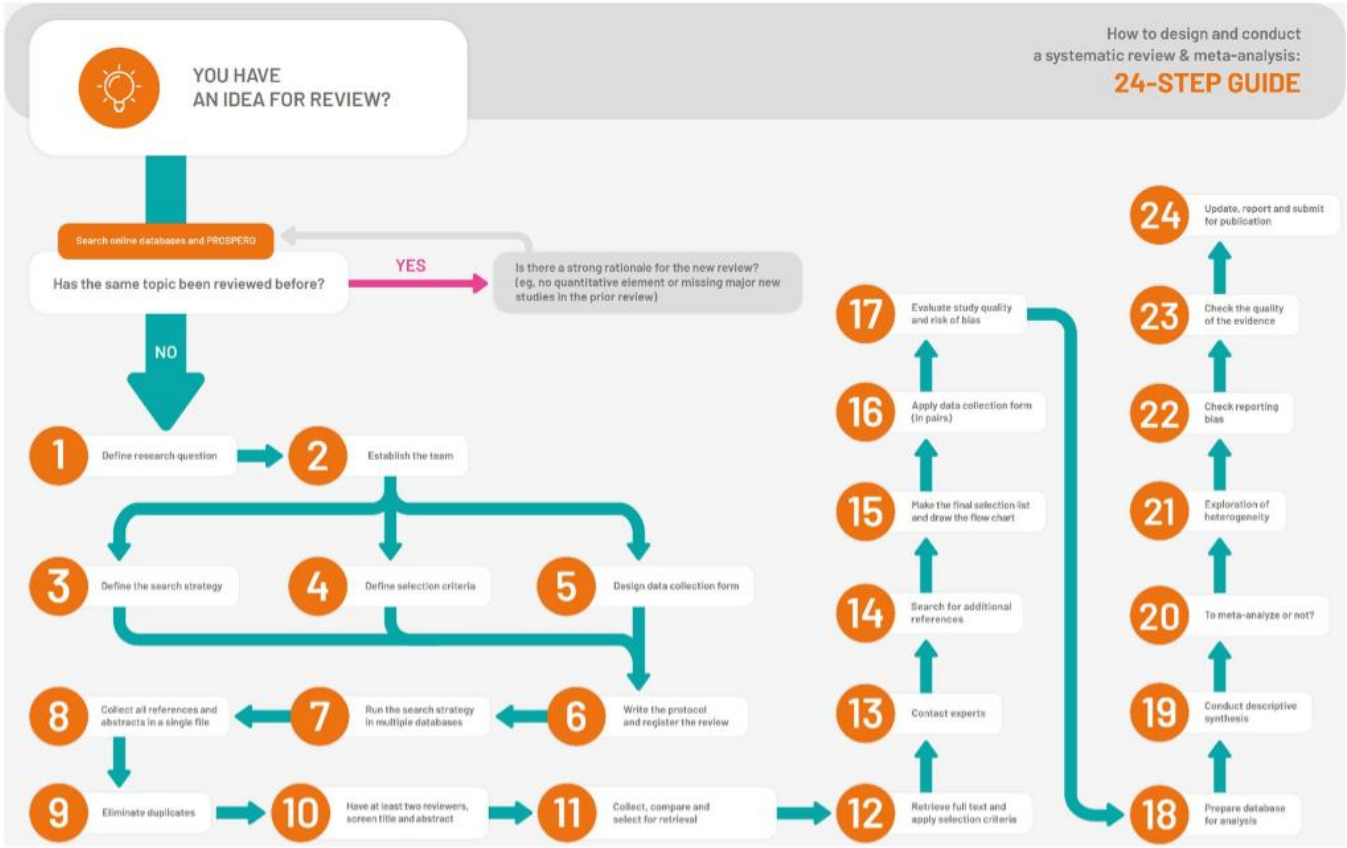
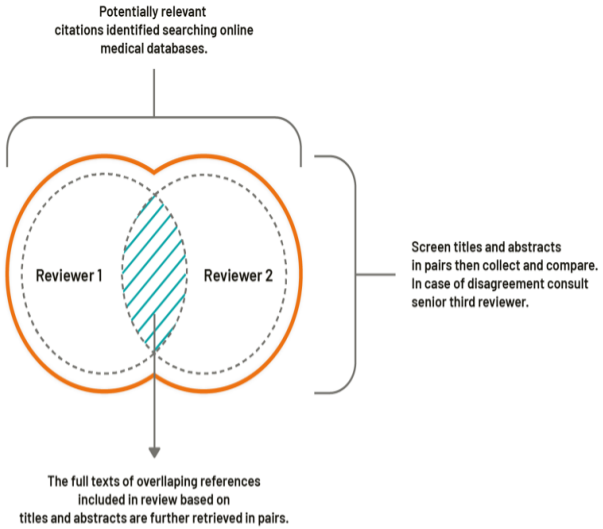



**Fig. 1** 24-STEP GUIDE checklist: steps to be followed to successfully design and conduct a systematic review and meta-analysis

**Fig. 2** Reference screening process



# *Thank You for your attention*

# *Any Questions*

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